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Data Organization and Visualization
5023-001
10/18/2024

An analysis of Temperature Homogeneity Issues within Temperature Recording and a practical example of it's importance

Introduction:

For the sake of overall public planning (infrastructure, zoning laws, agriculture, etc.) it is paramount to have up to date and accurate knowledge of weather patterns and heat indexes. Especially since, according to the literature, from January of 1895 to December of 2017, it was seen in all states aside from ten that temperatures rose approximately 2.9 degrees Celsius over approximately 100 years (Gil-Alana, Sauci, 2018). It is for this reason that various recordings, precipitation, humidity, and, the topic of this paper, temperature must be as uniform as possible in terms of collection and analysis in order to best plan for the future. Standardizing these various recordings is referred to as homogenizing them and has it's set of challenges, such as having to resort to certain, less favorable statistical analysis (Kruskal-Wallis, Pettitt, and Freidman tests) when relative tests (simply comparing results from similar testing) is not possible (Yozgatligil, Yazici, 2016). The need for such homogenization can be seen with many data sets from across the last one to one and a half centuries as the analysis that will follow illustrates. Over this time, enough changing of recording stations, surveying methods, and technological achievement and implementation has occurred to where even the most sophisticated measurement devices, such as satellites and climate models, have been shown to exaggerate or neglect key data just when implementing and modernizing the network (Lanzante, Klein, 2002). It is for this reason that some concerted, centralized effort needs to be made to standardize collection and analysis of not only current meteorological data, but investigation into past data as well.

Review of Literature:

As elucidated earlier, over the last 100 or so years temperature in the United States has risen approximately 2.9 degrees Celsius on average for the entire nation (Gil-Alana, Sauci, 2018) with the general trend being a negative correlation between the difference in maximum temperatures and minimum temperatures as each increase (Gil-Alana, 2017). The issue with this is that, for the previously described prerogative of using this data for public and economic planning, more descriptive and precise results are needed than what is being reported,

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especially when different data sets imply different statistical results. It's a well-known fact in the realm of meteorology and climate research that there are other factors that impact what goes into temperature reports than simply taking the temperature for the day, writing it down, and putting it into some kind of model. Even though there are several methods at researchers' disposal to try and standardize the data (relative testing, Kruskal-Wallis Testing, Pettitt testing, and Freidman Testing, all to various degrees of efficacy) (Yozgatligil, Yazici, 2016) actually standardizing the data presents numerous challenges that are not always accounted for in the final product.

This is best illustrated in a paper by Nordli et al, where an investigation was done into temperature recordings from the 1830's to the 2010's in Oslo, Norway. During said investigation, Nordi et al discovered that in the 1870's there was significant shielding of thermometers used in key temperature recording stations, leading to a marked decrease in recorded temperature (2015). The issue was ultimately rectified, after careful statistical analysis of the records to other records from similar stations, by adding 0.3 degrees Celsius to all recordings from the appropriate era. However, this illustrates a key fault in this kind of retroactive editing, as it fails to take into account various weather effects, such as cloud coverage, wind patterns, and precipitation, that all affect ultimate temperature. The effects of heat islands (the tendency for large municipalities to increase the temperature for weather stations)were taken into account, but even then the lack of clarification on how the aforementioned, non anthropogenic, factors impact the final temperature makes the manner of simply adding 0.3 degrees celsius a dubious tactic for homogenization. These types of homogenization issues are prevalent elsewhere, as Pandzic and Likso illustrated that, between 1960 and 2000, neglected and inferior weather stations in Croatia were off by less than an average of 0.5 degrees celsius for the allotted time frame (2010). When corrected using data from other stations the inhomogeneity in the data from the inferior stations was difficult to explain (2010). The paper also indicates that data exists within their investigation that seems to indicate that inhomogenization doesn't explain many of the temperature variance seen in the allotted record, results in accordance with other studies in Sweden in Switzerland were stated as corroborating this (2010). A study in Finland in Norway also had results indicating a drastic need for homogenization, as it reported that the kind of relocation and renovation common for weather stations altered temperature by plus or minus two degrees celsius (Tuomenvirta, 2000).

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As one can see, there are numerous examples of how homogenization presents serious challenges, some even where advanced technology doesn't alway solve the fundamental issues. Vast amounts of data has to be omitted in some situations to make the models remotely useful, such as how Lanzante and Klein had to delete a substantial 30% of previously recorded data to make their predictive model viable (2002). As one can see, serious efforts must be made towards making homogenization a more fruitful process, a fact which is even better corroborated by how there are clear nuances to the trends in temperature change. After homogenization, further evidence was accrued by Lee and Lund showing disparate changes in temperature throughout the United States over time (such as how areas of the South are cooling despite how the west, midwest, and New England are warming)(2013). The kind of temperature homogenization, as evidenced by Knapp, illustrated how, in some cases, temperature variations can be explained by pressure changes in the upper atmosphere (1992). Keiser and Griffiths, when analyzing temperature variations in the Northern Great Plains after homogenization, found that monthly maximum and minimum temperatures didn't fall in line with previous research using yearly maximums and means (1997).

As evidenced, simply comparing current measurements to past measurements is a fallacious paradigm when studying temperature differentials: some degree of homogenization must, and should occur, to derive useful insight into what the data is implying.

Methods:

An attempt was made to illustrate this need for better homogenization procedures was made by comparing data concerning the mean annual temperature in every United States county from 1895 to 2019 (henceforth referred to as the "core" data) with daily average taken in 1970 and 2019 from a separate data source (VisualCrossing.com, which will henceforth be known as the "auxiliary" data). Each set of data can be found in the following two links and submitted with this paper.

https://www.kaggle.com/datasets/thedevastator/analyzing-u-s-warming-rates-insights-into-climat

https://www.visualcrossing.com/?gad_source=1&gbraid=0AAAAAD5PI1gW87G41Z9B6iHEHt5h

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u1VS6&gclid=CjwKCAjwjsi4BhB5EiwAFAL0YEd54aHcnsB9Gg6CK-NkerFKDLtDtH7I8GleHR56zQAAAljl-w7xBxoC9WQQAvD_BwE

After which, the literature was collected and analyzed through python's nltk and PyPDF2 libraries in order to apply natural language processing to pick out key ideas and concepts from all the literature. Using Python, the core data was aggregated into yearly averages for each year and then the mean, along with all it's relevant values, by loading the two data sets into different data frames. Then, a pivot table was created for each where the state fips (a kind of reference code for counties) was used as the "index", the year for the "columns", and the temperature in degrees fahrenheit being used for the "values". Afterwards, the means were calculated for 1895, 1970, and 2019 using the "mean" function in python. The same process was done for both years in the auxiliary data where the index was set to the "datetime" column, and the values were set to the "temp." After that, the distributions for each data set were plotted using the matplot library and the seaborn library in order to best visualize how the overlap of the distributions (how hypothesis testing is often exemplified as in order to see how different the distributions have to be for rejecting the null hypothesis or not) looked to see how narrow or wide the rejection region is for each pairing of data.

Analysis:

After the 1895 data for the core data set was plotted, it was tested for normality via a Shairo-Wilks test. Afterwards, the same procedure was done to the yearly average for 2019 (in the same data set) and each was compared via standard hypothesis testing for a difference in mean. This was done amongst the core data for 1985 and 1970, and 1970 and 2019 (all still in the same core data set). After Which each was plotted in python using the matplotlib libraries and the seaborn libraries, with the degree of normality and the respective P-Values being calculated with the scipy.stats library. Then, normality was tested for the auxiliary for 1970 and 2019, after which (when they were each deemed abnormal) nonparametric testing was done via a Wilcoxon rank-sum test for a difference in the mean for each. The same testing was then done for the difference for the core 1970 and auxiliary 1970 data along with the core 2019 and auxiliary 2019 data, with the appropriate U-statistic and P-values being recorded for all three pairings. Afterwards, all the relevant P-values were used to corroborate or reject the null hypothesis for each hypothesis test for a difference in the means. The distributions for each data

set were also graphed with each appropriate pairing appearing on the same graph for easy visualization of how different procurement and analysis methods derive radically different results and also to best gauge how the overlap for the various testing looks to derive a rejection area.

Results and Discussion:

Natural Language Processing Results:

Topic 0: (0, '0.000*"," + 0.000*"(" + 0.000*"." + 0.000*")" + 0.000*"series" + 0.000*"time" + 0.000*"1" + 0.000*"trend" + 0.000*":" + 0.000*"The" + 0.000*"station" + 0.000*"change" + 0.000*"cid:0" + 0.000*"cid:1" + 0.000*"C" + 0.000*"–" + 0.000*";" + 0.000*"data" + 0.000*"temperature" + 0.000*"2"')

Topic 0 Sentence: 0.000*"," + 0.000*"(" + 0.000*"." + 0.000*")" + 0.000*"series" + 0.000*"time" + 0.000*"1" + 0.000*"trend" + 0.000*":" + 0.000*"the" + 0.000*"station" + 0.000*"change" + 0.000*"cid:0" + 0.000*"cid:1" + 0.000*"c" + 0.000*"–" + 0.000*";" + 0.000*"data" + 0.000*"temperature" + 0.000*"2".

Topic 1: (1, '0.000*"," + 0.000*")" + 0.000*"." + 0.000*"(" + 0.000*"series" + 0.000*":" + 0.000*"temperature" + 0.000*";" + 0.000*"The" + 0.000*"time" + 0.000*"1" + 0.000*"cid:0" + 0.000*"trend" + 0.000*"–" + 0.000*"mean" + 0.000*"station" + 0.000*"cid:1" + 0.000*"2" + 0.000*"data" + 0.000*"C"')

Topic 1 Sentence: 0.000*"," + 0.000*")" + 0.000*"." + 0.000*"(" + 0.000*"series" + 0.000*":" + 0.000*"temperature" + 0.000*";" + 0.000*"the" + 0.000*"time" + 0.000*"1" + 0.000*"cid:0" + 0.000*"trend" + 0.000*"–" + 0.000*"mean" + 0.000*"station" + 0.000*"cid:1" + 0.000*"2" + 0.000*"data" + 0.000*"c".

Topic 2: (2, '0.000*"," + 0.000*")" + 0.000*"(" + 0.000*"." + 0.000*":" + 0.000*"series" + 0.000*"temperature" + 0.000*";" + 0.000*"cid:0" + 0.000*"1" + 0.000*"–" + 0.000*"The" + 0.000*"trend" + 0.000*"station" + 0.000*"cid:1" + 0.000*"time" + 0.000*"data" + 0.000*"2" + 0.000*"%" + 0.000*"*"')

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Topic 2 Sentence: 0.000*"," + 0.000*")" + 0.000*"(" + 0.000*"." + 0.000*":" + 0.000*"series" + 0.000*"temperature" + 0.000*";" + 0.000*"cid:0" + 0.000*"1" + 0.000*"–" + 0.000*"the" + 0.000*"trend" + 0.000*"station" + 0.000*"cid:1" + 0.000*"time" + 0.000*"data" + 0.000*"2" + 0.000*"%" + 0.000*"'".

Topic 3: (3, '0.000*"," + 0.000*")" + 0.000*"." + 0.000*"(" + 0.000*":" + 0.000*"series" + 0.000*"cid:0" + 0.000*"1" + 0.000*"temperature" + 0.000*"trend" + 0.000*"–" + 0.000*";" + 0.000*"data" + 0.000*"station" + 0.000*"time" + 0.000*"cid:1" + 0.000*"The" + 0.000*"mean" + 0.000*"2" + 0.000*"'"')

Topic 3 Sentence: 0.000*"," + 0.000*")" + 0.000*"." + 0.000*"(" + 0.000*":" + 0.000*"series" + 0.000*"cid:0" + 0.000*"1" + 0.000*"temperature" + 0.000*"trend" + 0.000*"–" + 0.000*";" + 0.000*"data" + 0.000*"station" + 0.000*"time" + 0.000*"cid:1" + 0.000*"the" + 0.000*"mean" + 0.000*"2" + 0.000*"'".

Topic 4: (4, '0.074*"," + 0.058*")" + 0.057*"(" + 0.049*"." + 0.009*":" + 0.007*"series" + 0.006*"cid:0" + 0.006*"temperature" + 0.005*";" + 0.005*"1" + 0.005*"The" + 0.004*"–" + 0.004*"trend" + 0.004*"cid:1" + 0.004*"data" + 0.004*"time" + 0.004*"station" + 0.004*"2" + 0.003*"mean" + 0.003*"%"')

Topic 4 Sentence: 0.074*"," + 0.058*")" + 0.057*"(" + 0.049*"." + 0.009*":" + 0.007*"series" + 0.006*"cid:0" + 0.006*"temperature" + 0.005*";" + 0.005*"1" + 0.005*"the" + 0.004*"–" + 0.004*"trend" + 0.004*"cid:1" + 0.004*"data" + 0.004*"time" + 0.004*"station" + 0.004*"2" + 0.003*"mean" + 0.003*"%".

Topic Names (Themes): ['Topic 0', 'Topic 1', 'Topic 2', 'Topic 3', 'Topic 4']

It is evident from the Natural Language Processing that most of the topics centered around the general trend for each station across time changing in accordance to a changing mean of temperature and temperature series.

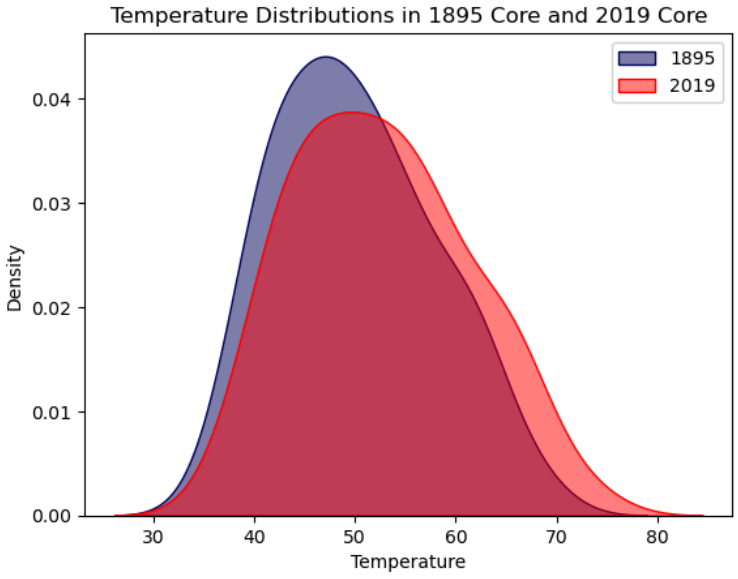
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Statistical Testing Results:

For the 1895 core data, a Shapiro-Wilks test for normality was done with the following results:
Percent Normal: 0.9656
P-Value: 0.1691
Conclusion: Data is approximately normal

For the 2019 core data, a Shapiro-Wilks test for normality was done with the following results:
Percent Normal: 0.9696
P-Value: 0.2448
Conclusion: Data is approximately normal

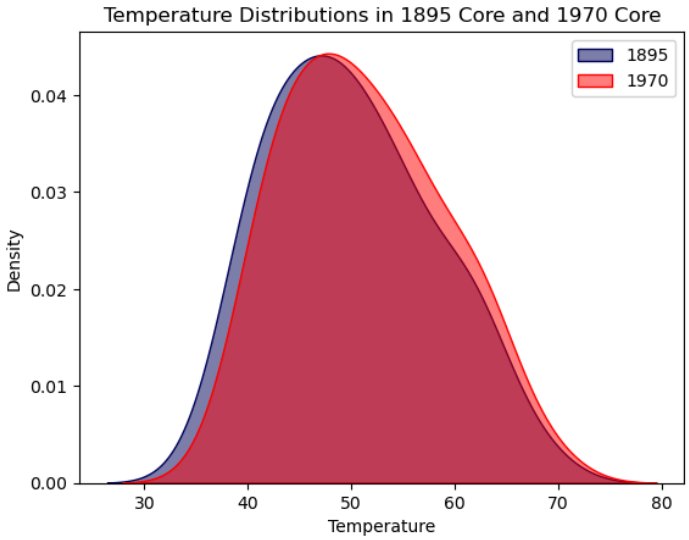
For the 1895 core and 2019 core pairing, a hypothesis test was set up wit the following parameters:
HO: (Mean of 1895 core) = (Mean of 2019 Core)
HA: (Mean of 1895 core) ≠ (Mean of 2019 Core)
P-Value: 0.1246
Conclusion: Fail to Reject the Null Hypothesis, the data shows no change in the mean between the two years



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For the 1970 core data, a Shapiro-Wilks test was performed for normality with the following results:
Percent Normal: 0.9680
P-Value: 0.2113
Conclusion: Data is approximately normal

For the 1895 core and 1970 core pairing, a hypothesis test was set up wit the following parameters:
HO: (Mean of 1895 core) = (Mean of 1970 Core)
HA: (Mean of 1895 core) ≠ (Mean of 1970 Core)
P-Value: 0.4427
Conclusion: Fail to Reject the Null Hypothesis, the data shows no change in the mean between the two years.



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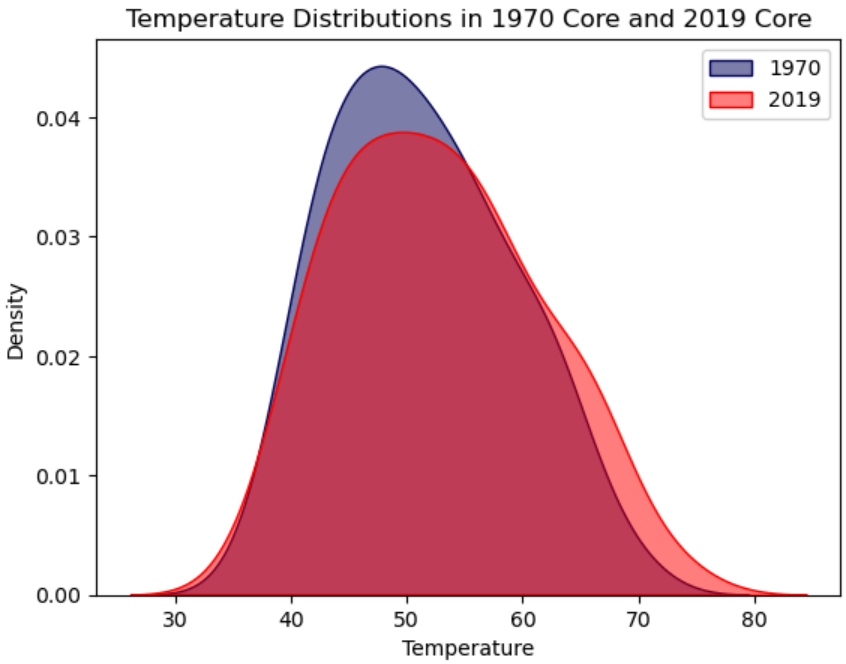
For the 1970 core and 2019 core pairing, a hypothesis test was set up wit the following parameters:

HO: (Mean of 1970 core) = (Mean of 2019 Core)

HA: (Mean of 1970 core) ≠ (Mean of 2019 Core)

P-Value: 0.4088

Conclusion: Fail to Reject the Null Hypothesis, the data shows no change in the mean between the two years.



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For the 1970 auxiliary and 2019 auxiliary pairing, a Wilcoxon rank-sum test was set up with the following parameters:

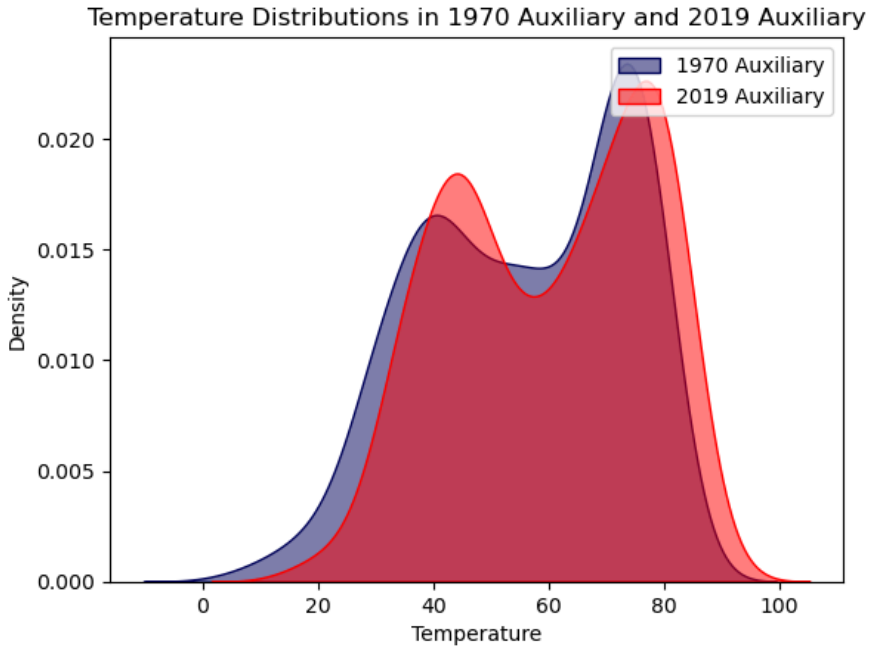
HO: (Mean of 1970 auxiliary) = (Mean of 2019 auxiliary)

HA: (Mean of 1970 auxiliary) ≠ (Mean of 2019 auxiliary)

U-Statistic: 57687.0

P-Value: 0.0012

Conclusion: Reject the Null Hypothesis and conclude that there is a significant difference between the 1970 auxiliary mean and the 2019 auxiliary mean



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For the 1970 core and 1970 auxiliary pairing, a Wilcoxon rank-sum test was set up with the following parameters:

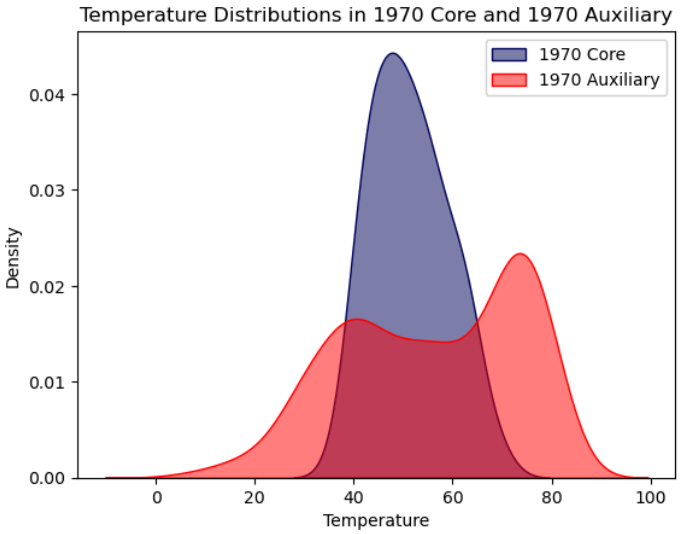
HO: (Mean of 1970 core) = (Mean of 1970 auxiliary)

HA: (Mean of 1970 core) ≠ (Mean of 1970 auxiliary)

U-Statistic: 7264.0

P-Value: 0.0512

Conclusion: Fail to reject the Null Hypothesis and conclude that there is not a significant difference between the 1970 core mean and the 1970 auxiliary mean



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For the 2019 core and 2019 auxiliary pairing, a Wilcoxon rank-sum test was set up with the following parameters:

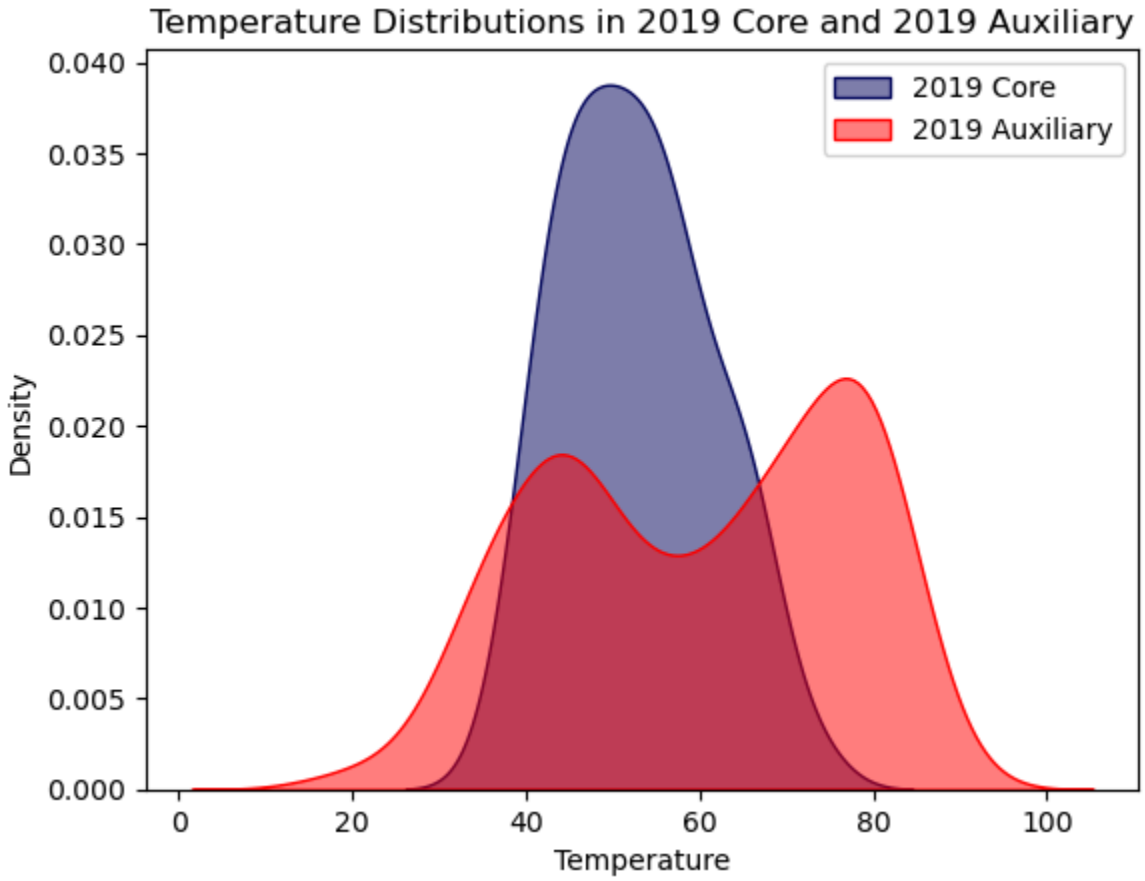
HO: (Mean of 2019 core) = (Mean of 2019 auxiliary)

HA: (Mean of 2019 core) ≠ (Mean of 2019 auxiliary)

U-Statistic: 6561.0

P-Value: 0.004353

Conclusion: Reject the Null Hypothesis and conclude that there is a significant difference between the 2019 core mean and the 2019 auxiliary mean



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Conclusion:

As one can see by the results of the statistical tests, each data set has remarkably different implications for temperature history and projections over time. The core data set, when one takes it's most extreme years of 1895 and 2019, corroborates how no significant temperature change has occurred throughout the United States as an aggregate. However, the auxiliary data indicates that there has been a substantial temperature change from 1970 to 2019. The results of the analysis also indicate that, for 1970, there was no substantial difference between the core and auxiliary data in terms of mean temperature, but there was one between the two for the 2019 mean temperature.

The implications of this study, despite it's limited scope in terms of data sets analyzed and the mysterious circumstances of the data (details such as collection methods, any homogenization, meteorological parameters, etc. are obfuscated), indicate that homogenization does have a necessary place in the realm of climate science and meteorological forecasting. If one were to simply account for one data set, he or she would either presume there is no direct evidence in temperature for any nationwide efforts to accommodate rising temperatures, or, would presume the opposite when there is evidence to the contrary. It is for this reason, and the reasons stated in the literature review section (i.e., how there are clearly unrefined aspects of how to perform homogenization and collect meaningful results from it such as with the research in Croatia and Oslo), that some effort needs to be applied to both applying homogenization to weather records throughout the time frame we have accurate records and make sure said homogenization efforts result in fruitful findings that allow deeper cognizance of the past trends in weather and their logical extrapolations.

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